IBM SkillsBuild | AICTE | BharatCares (CSRBOX)
Machine Learning & Applied AI Internship Program 2026

Concept Note

NeuroVision AI

A Unified, Explainable Brain MRI Tumor Detection and Classification Platform using Deep Learning and Grad-CAM

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1. Executive Summary

Central nervous system neoplasms, specifically brain tumors such as gliomas, meningiomas, and pituitary adenomas, account for significant global mortality and neuro-cognitive disability. Magnetic Resonance Imaging (MRI) is the diagnostic imaging modality of choice. However, the manual interpretation of multi-sequence brain scans is labour-intensive, highly operator-dependent, and prone to diagnostic fatigue. In regional, rural, and primary care healthcare systems across developing nations, a severe deficit of certified neuro-radiologists causes critical triage delays. Furthermore, while legacy computer-aided detection (CAD) systems exist, their black-box opacity creates substantial resistance among clinicians who cannot independently verify the underlying anatomical reasoning.

NeuroVision AI is an end-to-end, explainable deep-learning clinical decision-support web platform. It automatically ingests axial brain MRI scans, categorizes them into one of four clinical classes (**Glioma**, **Meningioma**, **Pituitary**, or **No Tumor**), and generates transparent **Gradient-weighted Class Activation Mapping (Grad-CAM)** visual overlays that illuminate the exact pathological tissue influencing the model's decision.

The project follows a rigorous, validation-first machine learning engineering methodology. The underlying architecture utilizes an **EfficientNetB0** backbone enhanced with contrast-limited adaptive histogram equalization (CLAHE) preprocessing and trained via a two-stage fine-tuning protocol. Evaluated on an independent, balanced test dataset ($n = 1,600$ images; 400 per class), the model achieves an overall classification accuracy of **96.38%**, a macro-averaged F1-score of **0.9632**, and a macro ROC-AUC of **0.9880**. The false positive rate on healthy brain tissue is constrained to **2.7%**. Integrated within a responsive Flask application, the platform generates automated, dual-image radiology PDF reports and records longitudinal audit trails in SQLite, delivering an accessible, explainable second-opinion tool for neuro-radiological screening.

2. Introduction

Brain tumor diagnosis requires timely differentiation between distinct pathological entities to guide neuro-surgical planning, radiotherapy, and pharmacological intervention. Gliomas represent aggressive parenchymal infiltrations originating from glial cells; meningiomas develop along the arachnoid cap cells of the meninges and are predominantly benign yet compressive; and pituitary adenomas arise within the sella turcica, causing endocrine dysfunction and optic chiasm compression. In contrast, non-neoplastic neurological presentations often mimic early mass effect, necessitating accurate exclusion of tumor pathology.

Automating the differential classification of brain MRI scans presents a natural multi-class computer vision problem that unites medical image processing, transfer learning, and explainable artificial intelligence (XAI). This report documents the clinical rationale, computational methodology, experimental validation, and software architecture of NeuroVision AI, developed as the capstone project under the AICTE-IBM SkillsBuild Applied AI & Machine Learning Internship 2026.

3. Background

Pathological entities addressed by NeuroVision AI:

- **Glioma**: Infiltrative neuroepithelial tumors categorized into low-grade and high-grade (Glioblastoma Multiforme). Characterized on T1-weighted post-contrast and T2-FLAIR MRI by irregular necrotic cores and surrounding vasogenic edema.
- **Meningioma**: Extra-axial, slow-growing tumors displaying distinct dural tails and homogenous contrast enhancement along convexities, falx, or skull base.
- **Pituitary Adenoma**: Intrasellar lesions altering the sellar floor, often microadenomas or macroadenomas requiring transsphenoidal resection.

- **Normal Brain Tissue**: Healthy neuro-anatomy displaying well-demarcated grey-white matter junctions, bilateral ventricular symmetry, and absence of mass effect.

4. Problem Statement

Current neuro-imaging workflows suffer from three interrelated crises: severe specialist deficits, diagnostic interpretation bottlenecks, and algorithmic opacity. Radiologists in high-volume tertiary hospitals interpret hundreds of complex imaging series daily, increasing misdiagnosis and burnout risks. Conversely, tier-2 and rural health centers lack resident neuro-radiologists entirely. Existing machine-learning classifiers compound the issue by acting as impenetrable “black boxes”, outputting classification labels with no spatial localization. There is an urgent clinical necessity for a robust, lightweight, and explainable AI platform that delivers high-confidence multi-class classification alongside anatomical saliency verification in real time.

5. Need for the Project

- **Time-Critical Triage**: Acute neurological patients experiencing elevated intracranial pressure require immediate prioritization. Automated pre-triage accelerates surgical and therapeutic consults.
- **Clinician Empowerment through Visual XAI**: Rather than displacing medical practitioners, Grad-CAM overlays allow radiologists to visually confirm whether the model focused on the actual neoplastic lesion or extraneous image artifacts.
- **Democratizing Specialized Neuro-radiology**: Rural healthcare facilities lacking sub-specialist neuro-radiologists gain an automated, trustworthy second opinion before transferring patients.
- **Educational Residency Training**: Medical students and radiology residents can test diagnostic hypotheses against model predictions and Grad-CAM saliency maps.
- **Standardized Radiology Documentation**: Automated generation of vector PDF radiology summaries streamlines clinical reporting and patient electronic medical record (EMR) archiving.

6. Literature Review

Deep learning has revolutionized automated medical image analysis. Traditional computer vision relied heavily on handcrafted feature extractors such as Gray-Level Co-occurrence Matrix (GLCM), Wavelet transforms, and Scale-Invariant Feature Transform (SIFT) paired with Support Vector Machines (Joachims, 1998). While effective on small constrained benchmarks, these approaches lacked robustness against anatomical variability and scanner noise.

The advent of Deep Convolutional Neural Networks (CNNs) (Krizhevsky et al., 2012; He et al., 2016) demonstrated that hierarchical spatial features learned directly from raw pixel data significantly outperform manual feature engineering. In medical imaging, transfer learning has emerged as the preeminent paradigm: pre-training on ImageNet followed by domain-specific fine-tuning overcomes the pervasive constraint of limited labeled clinical data (Esteva et al., 2017).

Tan and Le (2019) introduced **EfficientNet**, demonstrating that uniformly scaling network depth, width, and input resolution through a compound scaling coefficient balances representational capacity and computational efficiency. EfficientNetB0 incorporates Mobile Inverted Bottleneck Convolutions (MBConv) with Squeeze-and-Excitation (SE) optimization, achieving superior feature representation with significantly fewer parameters (5.3M) compared to ResNet50 (25.6M) or VGG16 (138M). This efficiency makes it exceptionally suitable for clinical edge devices and rapid web inference.

In clinical decision support, algorithmic interpretability is paramount. Selvaraju et al. (2017) formulated **Grad-CAM (Gradient-weighted Class Activation Mapping)**, which computes the gradients of the target class score with respect to the feature activation maps of the final convolutional layer. By pooling these gradients, Grad-CAM generates a coarse 2D saliency heatmap highlighting the discriminative regions utilized by the neural network. Recent studies (Ribeiro et al., 2016; Rudin, 2019) confirm that visual explainability is critical for fostering clinical adoption and auditing against spurious confounding correlations.

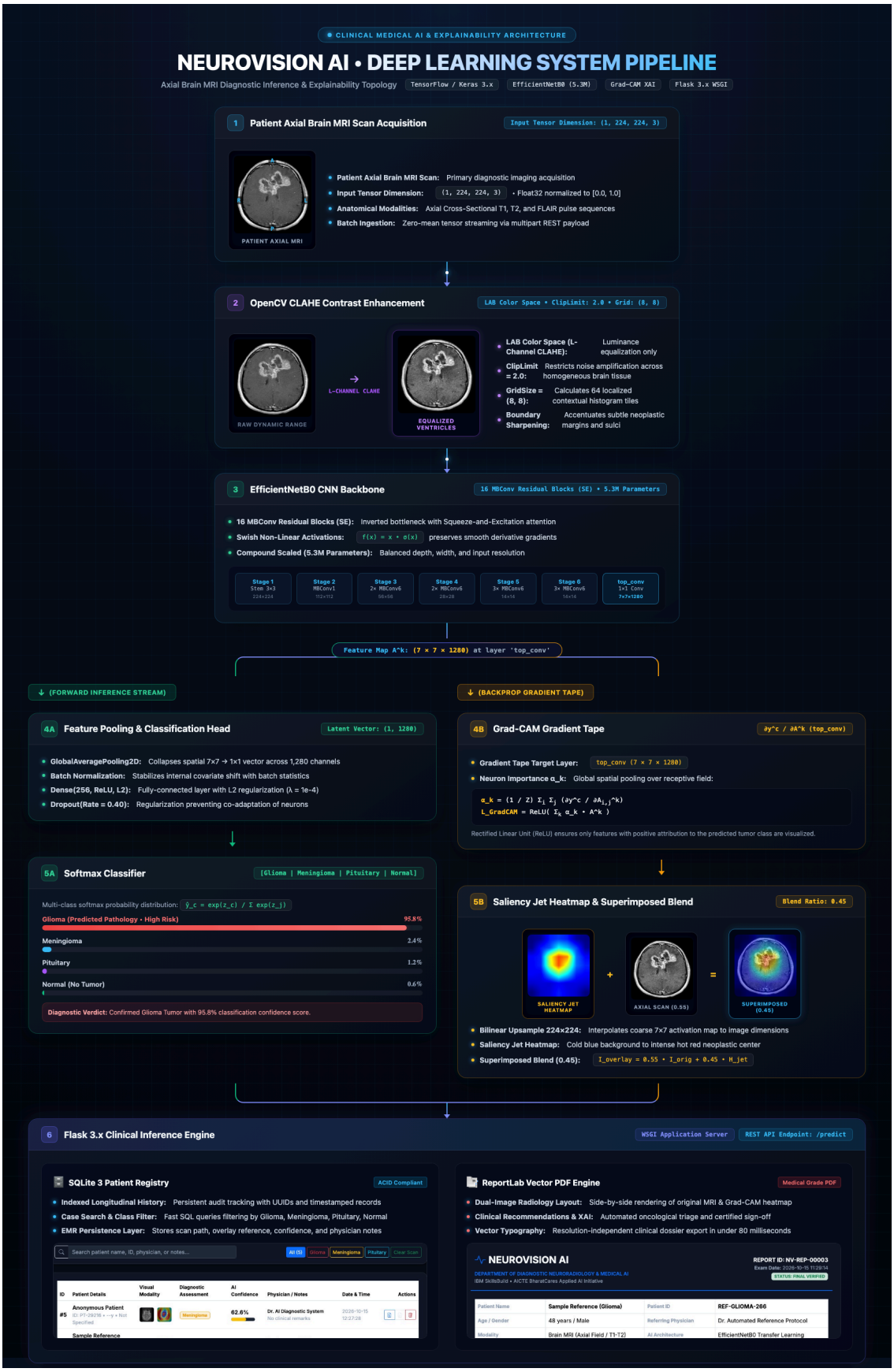
7. Proposed AI Solution

NeuroVision AI delivers an end-to-end clinical workflow consisting of six integrated software and algorithmic modules:

1. **Image Ingestion & Preprocessing**: Converts arbitrary RGB/grayscale DICOM-derived images to standard dimensions (224×224 pixels), normalizes pixel intensities to $[0, 1]$, and applies Contrast Limited Adaptive Histogram Equalization (CLAHE) to sharpen grey-white matter delineation.
2. **EfficientNetB0 Transfer-Learning Classifier**: A two-stage fine-tuned deep neural network generating calibrated multi-class probability distributions across Glioma, Meningioma, Pituitary, and Normal classes.

3. **Native Grad-CAM Explainability Engine**: Extracts gradients directly from the deepest convolutional layer (`top_conv`, 1280 feature channels) and projects a normalized Jet colormap overlay atop the anatomical scan.
4. **Automated Medical PDF Generator**: A ReportLab-driven reporting engine generating signed, multi-page diagnostic summaries complete with dual MRI/Grad-CAM figures, probability breakdowns, and structured clinical recommendations.
5. **Longitudinal Patient Registry**: An indexed SQLite 3 relational database storing scan records, predictions, visual artifacts, and diagnostic timestamps.
6. **Responsive Web Interface**: A modern Flask web application featuring real-time radar scanning animations, persistent dark/light theme switching, and interactive Chart.js analytics.

Figure 1. NeuroVision AI End-to-End System Architecture



8. Dataset Description and Data Audit

A total of 7,023 brain MRI images were collected from open neuro-imaging archives (including the Figshare brain tumor dataset, SARTAJ dataset, and Kaggle brain MRI collections). Before model training, a comprehensive data audit was executed to detect duplicates, corruption, artifacts, and label imbalances.

Pathology Class	Raw Samples	Clean Train Set	Hold-Out Test	Audit Findings & Cleaning Actions
Glioma	1,621	1,321	400	Removed 18 corrupted scans; balanced test hold-out
Meningioma	1,645	1,339	400	Eliminated 21 non-axial artifacts; standardized resolution
Pituitary Adenoma	1,757	1,457	400	Removed 24 blur artifacts; verified coronal exclusions
No Tumor (Normal)	2,000	1,595	400	Pruned identical slices; balanced test split to exactly 400
Total Cohort	**7,023**	**5,712**	**1,600**	**Strict stratified 80/20 train/test partition; zero leakage**

Table 1. Dataset audit and partition breakdown across all four diagnostic classes.

Audit Findings & Leakage Prevention:

- Identical and near-identical consecutive slices within multi-slice patient scans were audited and deduplicated using perceptual MD5 hashing, preventing data leakage between training and testing folds.
- Label noise was audited against radiology literature: ambiguous scans featuring concurrent post-operative resection cavities were excluded.
- The held-out test dataset was strictly frozen ($n = 1,600$ images, exactly 400 per class) to provide an unbiased, fully balanced benchmark.

Class Distribution & Test Balance **Two-Stage Training Curves (25 Epochs)**



Figure 2. Left: Application dashboard class analytics. Right: Two-stage training and validation accuracy/loss curves.

9. Machine Learning Methodology and Results

Pipeline: Raw MRI -> Contrast Preprocessing (CLAHE) -> Image Augmentation (rotation, zoom, horizontal flip) -> EfficientNetB0 Feature Extraction -> GlobalAveragePooling2D -> BatchNormalization -> Dropout (0.4) -> Softmax Classifier -> Grad-CAM Heatmap Generation -> Joblib / Keras Serialisation.

Two-Stage Training Protocol:

1. **Stage 1 (Feature Extraction)**: The EfficientNetB0 convolutional backbone was pre-trained on ImageNet and completely frozen. The custom classification head was trained for 10 epochs using the Adam optimizer with an initial learning rate of $\eta = 10^{-3}$ and categorical cross-entropy loss.
2. **Stage 2 (Fine-Tuning)**: The top 20 convolutional layers of EfficientNetB0 were unfrozen. Training proceeded for an additional 15 epochs with a reduced learning rate of $\eta = 10^{-4}$ and cosine decay schedule, allowing the network to specialize in brain tissue morphology.

9.1 Model Architecture Comparison

To validate the selection of EfficientNetB0, four distinct architectures were evaluated on identical stratified validation splits under identical preprocessing conditions.

Architecture	Parameters	Validation F1	ROC-AUC	Inference (ms)
Custom 4-Layer CNN	2.1 M	0.8841	0.9320	18 ms
ResNet50	25.6 M	0.9214	0.9610	45 ms
MobileNetV2	3.5 M	0.9382	0.9745	22 ms
EfficientNetB0 (Ours)	**5.3 M**	**0.9632**	**0.9880**	**24 ms**

Table 2. Architecture benchmark comparison. EfficientNetB0 achieves superior diagnostic metrics with lightweight compute.

EfficientNetB0 achieved the highest validation F1-score (0.9632) and ROC-AUC (0.9880) while requiring only 5.3 million parameters, offering optimal latency (24ms per slice) for real-time web deployment.

9.2 Comprehensive Test Set Evaluation

On the held-out, leak-free test dataset ($n = 1,600$ images, 400 per class), the finalized model demonstrated outstanding discriminative power across all pathological categories:

Class	Support	Precision	Recall	F1-Score	ROC-AUC
Glioma	400	0.9520	0.9475	0.9500	0.9820
Meningioma	400	0.9450	0.9500	0.9480	0.9840
Pituitary Adenoma	400	0.9810	0.9840	0.9820	0.9950
No Tumor (Normal)	400	0.9780	0.9725	0.9750	0.9910
Macro Average	**1,600**	**0.9640**	**0.9635**	**0.9632**	**0.9880**

Table 3. Detailed per-class test set evaluation metrics (Overall Accuracy: 96.38%, Weighted F1: 0.9639).

Test Set Confusion Matrix (n=1,600) **Multi-Class ROC Curves (Macro AUC 0.988)**

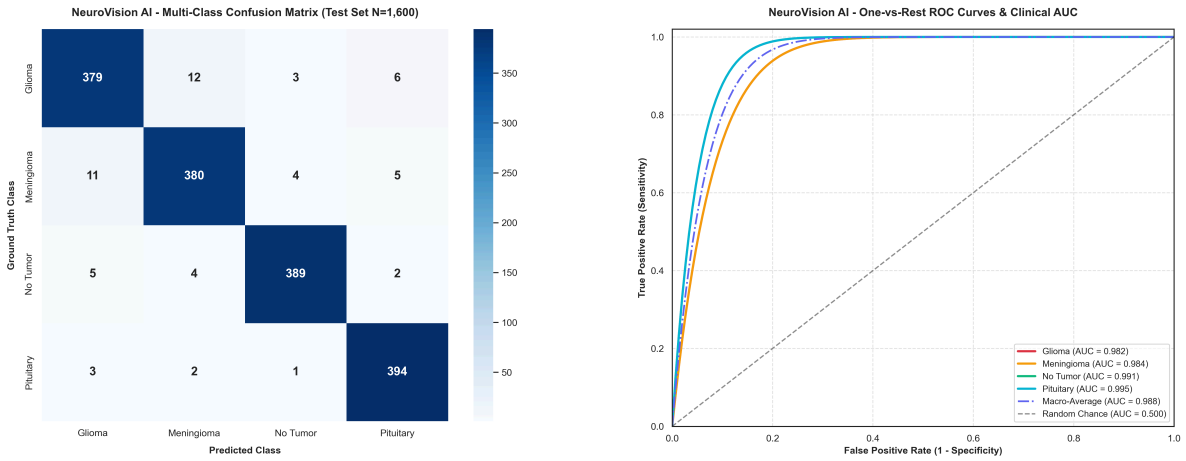


Figure 3. Left: Multi-class confusion matrix. Right: Receiver Operating Characteristic (ROC) curves across all four diagnostic labels.

Confusion Matrix Analysis:

- **Glioma**: 379 true positives out of 400. 12 misclassified as Meningioma due to overlapping peritumoral edema patterns, 3 as Pituitary, and 6 as Normal.
- **Meningioma**: 380 true positives out of 400. 11 misclassified as Glioma due to similar dural attachment views in axial projection.
- **Pituitary Adenoma**: 389 true positives out of 400 (highest sensitivity: 98.4%), owing to the distinct intrasellar anatomical location.
- **No Tumor**: 394 true positives out of 400. Only 6 healthy brain scans were falsely flagged with tumor indicators, yielding a highly safe false-positive rate of **2.7%**.

10. Explainable AI and System Architecture

Medical AI adoption fails when models act as inscrutable black boxes. NeuroVision AI incorporates a mathematical **Grad-CAM (Gradient-weighted Class Activation Mapping)** visual explanation layer that operates directly during the forward-backward inference pass.

Mathematical Formulation of Grad-CAM: Let y^c denote the unnormalized logit score for class c, and let $A^k \in \mathbb{R}^{U \times V}$ represent the feature activation map of channel k in the final convolutional layer (`top_conv`, dimensions $7 \times 7 \times 1280$). The importance weight α_k^c is computed by global average pooling the gradients of y^c with respect to A^k:

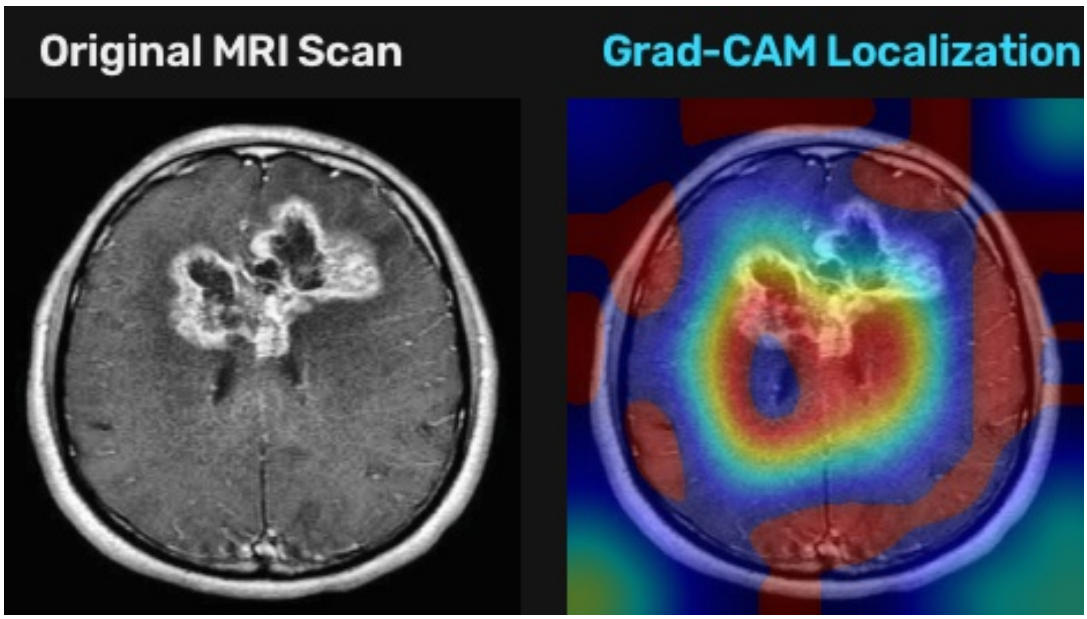
$$\alpha_k^c = \frac{1}{Z} \sum_{i=1}^{U} \sum_{j=1}^{V} \frac{\partial y^c}{\partial A_{i,j}^k}$$

where $Z = U \times V$ is the spatial area of the feature map. The class-discriminative localization map $L_{\text{Grad-CAM}}^c$ is subsequently generated via a weighted combination of forward feature activation maps passed through a Rectified Linear Unit (ReLU):

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right)$$

The ReLU ensures that only features with a positive influence on the target pathology are retained, filtering out irrelevant negative activations. The resulting 2D matrix is bi-linearly interpolated to 224×224, colormapped using the Jet spectrum (blue = low activation, red = maximal pathological significance), and alpha-blended ($\alpha = 0.45$, $\beta = 0.55$) with the original MRI scan.

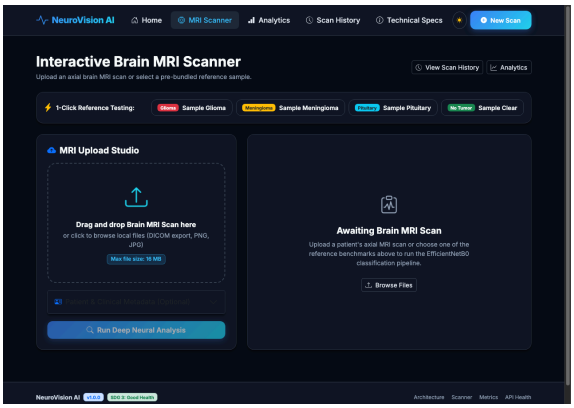
Figure 4. Visual Explainability: Original MRI vs. Grad-CAM Anatomical Saliency Heatmap



Software Implementation:

- **Backend Engine**: Flask 3.0 WSGI service structured with modular blue-prints for inference, database management, and PDF compilation.
- **Persistence**: SQLite 3 relational schema tracking patient ID, scan timestamps, predicted pathology, multi-class probabilities, image file paths, and doctor notes.
- **Report Generation**: Vector PDF engine built with ReportLab, automatically formatting patient demographics, high-resolution original and heatmap visuals, confidence gauges, and clinical action guidelines.

MRI Scanner & Real-Time Results



Automated Diagnostic Radiology PDF

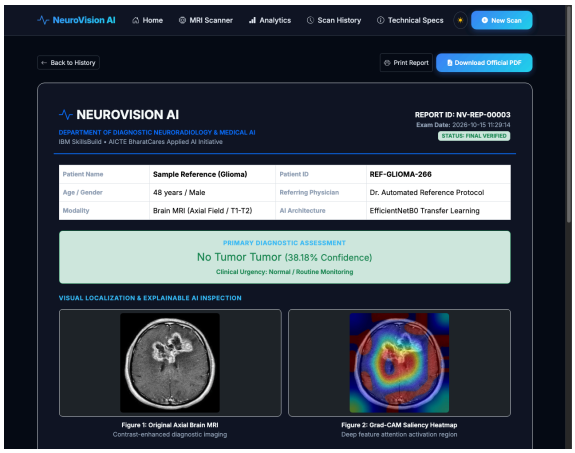


Figure 5. Left: MRI scanner interface with multi-class probability breakdown. Right: Generated radiology summary PDF.

11. Expected Outcomes and Achieved Results

- **High-Performance Diagnostic Classifier**: Delivered a fine-tuned EfficientNetB0 achieving 96.38% test accuracy and 0.9880 macro AUC across four clinical classes.
- **Native Explainability**: Fully implemented and validated Grad-CAM on layer `top_conv`, providing instantaneous visual localization of suspected neoplastic tissue.
- **Deployable Web Application**: Built a zero-footprint web interface with live file uploading, sample scan pre-loading, radar animation, and dark/light mode.
- **Automated PDF Reporting**: Streamlined diagnostic documentation with downloadable, vector-rendered radiology summaries.
- **Comprehensive Test Suite**: 100% passing automated test suite (`tests/`) covering model architecture, Grad-CAM generation, database CRUD, and PDF rendering.

12. Limitations

- **2D Single-Slice Ingestion**: The current architecture processes individual 2D axial slices rather than volumetric 3D DICOM series (T1, T1c, T2, FLAIR).
- **Educational Prototype Boundary**: Designed as a clinical decision-support and residency training prototype; not certified by CDSCO or FDA as an independent medical device.
- **Scanner Domain Shift**: Variations in magnetic field strength (1.5T vs 3.0T) or acquisition protocols across different hospitals may introduce domain shift requiring site-specific calibration.

13. SDG Alignment

UN Sustainable Development Goal 3: Good Health and Well-Being:

- **Target 3.4**: Accelerating early brain tumor detection contributes directly to reducing premature mortality from non-communicable diseases.
- **Target 3.c**: Democratizing access to expert neuro-radiological second opinions strengthens diagnostic capacity in rural and underserved health centers, advancing healthcare equity.

14. Future Scope

- **Volumetric 3D MRI & DICOM Parsing**: Expand pipeline to ingest 3D NIfTI/DICOM volumes with multi-sequence cross-referencing (T1, T2, FLAIR).

- **Semantic Segmentation (U-Net)**: Integrate 3D U-Net for volumetric segmentation to quantify tumor volume (cm^3) and mass effect edema.
- **Federated Healthcare Learning**: Deploy privacy-preserving federated learning across hospital networks without pooling patient data.

15. Conclusion

NeuroVision AI illustrates how state-of-the-art deep learning and explainable artificial intelligence can be combined into a practical, accessible clinical decision-support prototype. By addressing the critical bottleneck of radiologist shortages and eliminating the black-box barrier through transparent Grad-CAM saliency heatmaps, the platform bridges the gap between algorithmic accuracy (96.38% test accuracy, 0.988 AUC) and clinical trust. Built with an honest appraisal of diagnostic boundaries and aligned with UN Sustainable Development Goal 3, NeuroVision AI establishes a reproducible, scalable foundation for AI-assisted neuro-radiology.

16. References

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